

§3 Partial Fractions

* Let $P(x)$ and $Q(x)$ be polynomials.

If $P(x) = Q(x)S(x) + R(x)$ where $\deg Q(x) > \deg R(x)$, then

$$\begin{aligned}\int \frac{P(x)}{Q(x)} dx &= \int \frac{Q(x)S(x) + R(x)}{Q(x)} dx \\ &= \int \left[S(x) + \frac{R(x)}{Q(x)} \right] dx\end{aligned}$$

Exam 1 Find $\int \frac{x^3 + x}{x - 1} dx$.

(sol)

Case I The denominator $Q(x)$ is a product of distinct linear factors.

Let $Q(x) = (a_1x + b_1)(a_2x + b_2) \cdots (a_nx + b_n)$.

There exist constants $k_1, k_2, \dots, k_n \in \mathbb{R}$ such that

$$\frac{R(x)}{Q(x)} = \frac{k_1}{a_1x + b_1} + \frac{k_2}{a_2x + b_2} + \cdots + \frac{k_n}{a_nx + b_n}.$$

Exam 2 Evaluate $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$.

(sol)

Case II $Q(x)$ is a product of linear factors, some of which are repeated.

Let $Q(x) = (a_1x + b_1)^r (a_2x + b_2) \cdots (a_nx + b_n)$.

There exist constants $k_{11}, k_{12}, \dots, k_{1r}, k_2, \dots, k_n \in \mathbb{R}$ such that

$$\frac{R(x)}{Q(x)} = \frac{k_{11}}{a_1x + b_1} + \frac{k_{12}}{(a_1x + b_1)^2} + \dots + \frac{k_{1r}}{(a_1x + b_1)^r} + \frac{k_2}{a_2x + b_2} + \dots + \frac{k_n}{a_nx + b_n}.$$

Exam 3 Find $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$.

(sol)

Case III $Q(x)$ contains irreducible quadratic factors, none of which is repeated.

Let $Q(x) = (ax+b)(cx+d)(ex^2+fx+g)(hx^2+ix+j)$.

There exist constants $k_1, k_2, \dots, k_6 \in \mathbb{R}$ such that

$$\frac{R(x)}{Q(x)} = \frac{k_1}{ax+b} + \frac{k_2}{cx+d} + \frac{k_3x+k_4}{ex^2+fx+g} + \frac{k_5x+k_6}{hx^2+ix+j}.$$

(cf) $\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$

Exam 4 Evaluate $\int \frac{2x^2-x+4}{x^3+4x} dx$.

(sol)

Exam 5 Evaluate $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$.

(sol)

Case IV $Q(x)$ contains a repeated irreducible quadratic factor.

Let $Q(x) = (ax + b)(cx^2 + dx + e)^r$.

There exist constants $k_1, k_2, \dots, k_{2r+1} \in \mathbb{R}$ such that

$$\frac{R(x)}{Q(x)} = \frac{k_1}{ax + b} + \frac{k_2x + k_3}{cx^2 + dx + e} + \frac{k_4x + k_5}{(cx^2 + dx + e)^2} + \dots + \frac{k_{2r}x + k_{2r+1}}{(cx^2 + dx + e)^r}.$$

Exam 6 Write out the form of the partial fraction decomposition of the function

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3}.$$

(sol)

Exam 7 Evaluate $\int \frac{1-x+2x^2-x^3}{x(x^2+1)^2} dx$.

(sol)

(Exam) Prove the recurrence relation

$$\int \frac{dx}{(x^2 + a^2)^n} = \frac{1}{2(n-1)a^2} \left[\frac{x}{(x^2 + a^2)^{n-1}} + (2n-3) \int \frac{dx}{(x^2 + a^2)^{n-1}} \right]$$

where $n \geq 2$.

(pf)

(Exam)

$$\int \frac{dx}{(x^2 + 2^2)^3} =$$